Shane Chu

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Professional Summary

PhD-trained researcher focused on motif discovery and their functional effects in biological sequences. Core skills span deep learning interpretation, randomized algorithms, and high-performance computing—from proving approximation guarantees for attribution methods to writing custom CUDA kernels for large-scale optimization. Seven years building methods that extract combinatorial patterns from large-scale datasets.

Professional Experience

Postdoctoral Researcher Jan 2024 – Present

Columbia University *New York, NY*

- **Learned Attributions:** Proved that additive decomposition unifies modern interpretation methods (SHAP, DeepLIFT, Integrated Gradients); trained an auxiliary neural network that produces per-feature attributions in a single forward pass with provable approximation bounds on power indices.
- **Combinatorial Pattern Discovery:** Designed a GPU-parallelized count-min-sketch for discovering higher-order feature combinations (up to 5-way) with provable frequency guarantees; applied cooperative game theory to quantify non-additive interaction effects via regression on Shapley values.
- **GPU-Accelerated Bayesian Optimization:** Implemented Bayesian optimization with custom CUDA kernels using Nyström approximation and randomized SVD, scaling to 200K candidate sequences.
- **PPR Binding Map Acceleration:** Reimplemented a protein–RNA binding site mapping method (PPRdecoder), achieving 150× speedup through efficient data structures and algorithmic improvements.

Doctoral Researcher Sep 2018 – Dec 2023

Washington University in St. Louis *St. Louis, MO*

- **Sparse Representation Framework:** Developed a sparse representation framework (MOTIFs.jl) for pattern discovery in sequential data, outperforming traditional methods on benchmark datasets.
- **Statistical Pattern Discovery Tools:** Built SeqShuffle.jl (k-mer preserving sequence shuffling) and MotifPvalue.jl (efficient p-value computation), enabling rigorous statistical inference at scale.

ML Research Intern May 2023 – Aug 2023

Regeneron Pharmaceuticals *Tarrytown, NY*

- **Bayesian Optimization Pipeline:** Applied Bayesian ML and metaheuristics for promoter design in a discrete, high-dimensional search space with expensive evaluation costs.
- **Comparative Analysis Pipeline:** Built a Snakemake pipeline to identify conserved promoter elements across primates utilizing JASPAR, MOODS, and phylogenetic conservation scores.

Research Assistant Nov 2017 – May 2018

Center for Remote Sensing of Ice Sheets *Lawrence, KS*

- **HPC & model selection:** Built scalable optimization pipeline integrating hyperparameter tuning and model selection for ice detection from radar imagery.

Technical Skills

Programming & Systems

- **Languages:** Python, Julia, C/C++, R, MATLAB, SQL
- **GPU & HPC:** CUDA kernel development, kernel fusion, custom memory allocation, distributed computing (Slurm, DDP)
- **Infrastructure:** Docker, Snakemake, Git, system design

Machine Learning & Mathematics

- **ML:** PyTorch, Flux.jl; custom backward passes, interpretability (Shapley values, integrated gradients), diffusion models
- **Optimization:** Convex optimization, Bayesian optimization, operator splitting (ADMM), metaheuristics
- **Probability & Statistics:** Concentration inequalities, Bayesian nonparametrics, Gaussian processes
- **Algorithms:** Randomized algorithms, hashing, streaming data structures, dynamic programming

Selected Work

- Chu, S. K., & Zhang, C. (2026). Identifying motif combinations with extreme expression (In preparation).
- Chu, S. K., & Zhang, C. (2026). Mutation logos reveal substitution motifs at the extremes of deep mutational scans (In preparation).
- Chu, S. K., & Stormo, G. D. (2023). Finding motifs using DNA images derived from sparse representations. Bioinformatics, 39(6), btad378.
- Paden, J. D., Berger, V., Al-Ibadi, M., Chu, S. K., Xu, M., Crandall, D., & Fox, G. (2018, December). Subglacial bed topography using machine learning and geostatistical analysis applied to 2D and 3D radar sounding. In AGU Fall Meeting Abstracts (Vol. 2018, pp. C43A-07).
- Berger, V., Xu, M., Chu, S. K., Crandall, D., Paden, J., & Fox, G. C. (2018, July). Automated tracking of 2D and 3D ice radar imagery using VITERBI and TRW-S. In IGARSS 2018-2018 IEEE International Geoscience and Remote Sensing Symposium (pp. 4162-4165). IEEE.

Talks & Posters

- Poster, Identifying motif combinations with extreme expression, AI in Biology, Cold Spring Harbor Laboratory May 2026
- Motif discovery and combinatorial effects via learned attributions, Systems Biology Seminar, Columbia University Jan 2025
- DNA motifs revealed through sparse representations, Systems Biology Initiative, Columbia University Oct 2024
- Poster, Machine Learning in Computational Biology, ISMB Conference Jul 2022

Open Source Contributions

ATLAS: GPU-accelerated Julia library for discovering co-occurring motif combinations that drive expression, via learned attributions.
EpicHyperSketch.jl: Parallel count-min-sketch for counting in fixed-size space on the GPU.
AutoComputationalGraphTuning.jl: Automatic hyperparameter tuning for computational graphs.
MOTIFs.jl: Sparse representation-based pattern discovery for sequential data.
EntroPlots.jl: Information-theoretic visualization library for pattern analysis.
RealLabelNormalization.jl: Label normalization utilities for regression tasks.
SEQ2EXPdata.jl: Efficient data structures for sequence-to-scalar prediction tasks.
SeqShuffle.jl: K-mer preserving string permutation for null model generation.
MotifPvalue.jl: Efficient p-value computation via dynamic programming approximation.

Education

Washington University in St. Louis St. Louis, MO
Ph.D. in Computer Science *Aug 2018 – Dec 2023*

- Focus: Machine Learning, Optimization, Algorithm Design
- Thesis: Interpretable deep learning via sparse representations for sequential pattern discovery
- Advisor: Gary D. Stormo

University of Kansas Lawrence, KS
Bachelor of Arts in Mathematics *Aug 2014 – May 2018*

Teaching & Leadership

Assistant Instructor Jan 2021 – Jun 2022
Washington University in St. Louis *St. Louis, MO*

- Designed lecture notes on variational inference and expectation maximization for Bayesian ML (CSE 515T) and mathematical optimization (ESE 415).
- Mentored 80+ students in optimization and advanced ML concepts, both theory and implementation; voted highest-rated teaching assistant for the optimization course.

Languages

- English (Fluent) and Mandarin (Native).